

CHEMICAL ANALYSIS – TESTS FOR GASES

KEY VOCABULARY

Hydrogen test: lit splint gives a squeaky pop.

Oxygen test: relights a glowing splint.

Carbon dioxide test: turns limewater milky/cloudy.

Chlorine test: bleaches damp blue litmus paper white.

COMMON MISCONCEPTIONS

~~—Oxygen pops with a lit splint.~~

✓ Oxygen relights a **GLOWING** splint; hydrogen pops.

~~—CO₂ turns limewater clear.~~

✓ CO₂ turns limewater cloudy (milky).

~~—Chlorine turns litmus red permanently.~~

✓ Chlorine bleaches the paper white.

GAS TEST SUMMARY

Hydrogen → lit splint → squeaky pop.

Oxygen → glowing splint → relights.

Carbon dioxide → limewater → goes cloudy.

Chlorine → damp litmus → bleached white.

Always state the test AND the result.

1

HYDROGEN

Describe the test and result for hydrogen. [2 marks]

2

OXYGEN

Describe the test and result for oxygen. [2 marks]

3

CARBON DIOXIDE

Describe the test and result for carbon dioxide. [2 marks]

4

CHLORINE

Describe the test and result for chlorine. [2 marks]

5

FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

A glowing splint ____ in oxygen; carbon dioxide turns limewater ____.

pop

relights

cloudy

bleaches

6

IDENTIFY

A gas relights a glowing splint. Identify the gas. [1 mark]